

MASTER PROJECT AUDIT — BIST ELITE AI ****Audit:**** A-001 MASTER PROJECT AUDIT ****Date:**** 2026-08-08
****Auditor:**** Principal AI Developer (read-only; no implementation performed) ****Repo:****
C:\Users\aliul\Desktop\bist-elite-ai (github.com/aliulu0/bist-elite-ai)

1. PROJECT VERDICT ****BIST ELITE AI is an "AI-Powered Early Opportunity Detection Platform" for Borsa Istanbul — NOT a stock screener.**** The core vision (Early Opportunity Detection via 11-engine composition) is ****delivered and working (GREEN BUILD, GREEN TESTS for the core)****. The remaining work is data pipeline completion, ML, hardening, and go-live.

Dimension	Score
Production readiness	**72%**
Feature completion	**70%**
Code quality	**72%**
Architecture quality	**85%**
Reuse quality	**75%**
Overall	**74 / 100 (B)**

2. HEADLINE NUMBERS

Metric	Count
Total files (TS/TSX/PY/JSON/JS)	~1,985 (1,367 TS + 489 TSX + 25 PY + 96 JSON + 8 JS)
Total LOC (TS + TSX + PY)	~204,862
Modules (NestJS)	~78
Controllers	48 (47 + root health)
Services	225
Engines	39 (19 core + 20 supporting)
Registries	~15
DTOs	58
API endpoints	~150+
Dashboards	10 surfaces (canonical web + legacy web + telegram)
Scheduler jobs	17
WebSocket events	8
Event-bus categories	9

Provider adapters	9 real (Yahoo, Fintables, Finnhub, AlphaVantage, SerpAPI, Google News, KAP, TCMB, MKK) + AgentReach
Test files	~347 (306 API spec + 13 web + 19 legacy + 6 python + 3 e2e)
Tests (approx)	~4,100–4,500
Failing tests	5 (3 suites, deterministic)
Git commits	5

3. ENGINE VERDICTS

Engine	Verdict
Prediction	Production Ready
Research Hub	Production Ready
Verification AI	Production Ready
Catalyst	Production Ready
Smart Money	Production Ready
Elite Score	Production Ready (3 modules to consolidate)
Opportunity	Production Ready
Decision	Production Ready
Entry Zone	Production Ready
Backtest	Needs Improvement (unused RSI + duplication)
Early Opportunity (CORE)	**Production Ready — 11/11 chain links verified**
Multi-Timeframe	Production Ready
Financial Rules	Production Ready
Indicator	Production Ready (single source of truth, D004)
Market Structure	Production Ready
Macro	Needs Improvement (`Math.random()` demo signal)
Portfolio AI	Needs Improvement (frontend demo data)
Learning (adaptive/regime)	Production Ready (rule-based)
Self Learning	Prototype (in-memory, not scheduled)

Missing engines: none. **Placeholders:** Self-Learning (prototype), Market Overview (partial stub), Watchlist (stub), Worker (health stub).

4. DATA LAYER VERDICT

- **9 real HTTP providers** verified (no production mocks; mocks confined to ``.spec.ts``).
- **TradingView provider MISSING** (D002 priority #2) — biggest data gap.
- **QualityScorer bug:** ``new Date(c.provider)`` → NaN → staleness always scores 50.
- Dual stack (legacy Yahoo service vs unified orchestrator) — D012 migration partial.
- No provider API keys configured; only Yahoo/GoogleNews/KAP work keyless.

5. GITHUB INTEGRATION VERDICT ****None of the 5 referenced repositories are code-integrated.**** NoFx (UI aesthetic), tradingagents (R3 roadmap), vectorbt (metrics concept, native TS), agent-reach (name coincidence), ai-berkshire (R4 roadmap). No code copied/vendored. Highest-value future integrations: vectorbt on worker stub, TradingAgents multi-agent, AI Berkshire.

6. CRITICAL FINDINGS (must fix)

ID	Finding
C1	5 duplicate <code>`@Controller`</code> prefixes → route collisions (portfolio, watchlist, scanner, dashboard, analysis)
C2	2 stub controllers (watchlist in-memory, market-overview mock)
C3	1 orphaned controller (market-scanner) — dead code
C4	<code>`deploy.yml`</code> uses nonexistent <code>`docker/build-and-push-action@v6`</code> — CI deploy would fail
C5	QualityScorer staleness bug always scores 50
C6	Legacy <code>`frontend/`</code> dashboard page won't compile (broken imports)
C7	5 deterministic failing tests ship with every release
C8	Orphaned e2e specs (203 tests) not wired into CI

7. BIGGEST RISKS 1. ****Route collisions**** (C1) — could serve wrong data / security-adjacent confusion. 2. ****Uncommitted bulk**** — 2,108 changed + 15,044 untracked files; a single bad reset loses the entire R2-026/27/28/29 + data-layer work. 3. ****No enforced coverage thresholds + failing tests**** — quality can silently regress. 4. ****No real deployment**** — `GO_LIVE_CHECKLIST` fully unchecked; HTTPS not configured. 5. ****Dual frontend + dual data stack**** — drift risk compounds over time.

8. TOP PRIORITIES 1. Fix 5 failing tests + wire orphaned e2e specs + enforce coverage thresholds. 2. Resolve 5 duplicate controller prefixes + register/delete orphaned controller. 3. Fix QualityScorer bug; add TradingView provider. 4. Replace 2 stub controllers with DB-backed implementations. 5. Wire scheduler job for early-opportunity self-learning. 6. Remove ``Math.random()`` from Macro; fix backtest RSI. 7. Fix deploy action + version drift; enable HTTPS. 8. Consolidate legacy ``frontend/`` vs ``apps/web``; fix broken dashboard route.

9. PROMPT & TIME ESTIMATE

- **Completed prompts:** R2-001 ... R2-029 (plus foundation/sprints 1-12) → ~29 roadmap prompts done.
- **Remaining roadmap prompts:** R3-001 (vectorbt/TS vectorized), R3-002 (Python quant), R3-003 (TradingAgents), R3-004 (Agency Agents), R4 (long-term vision: AI Berkshire, portfolio manager, risk manager, strategy builder,

learning engine) → **~8-10 roadmap prompts**.

- **New required prompts (gap register G1-G30):** ~20 additional implementation prompts.
- **Total remaining: ~28-32 prompts** → estimated **6-9 weeks** (1 prompt ≈ 1-2 focused days).
- **Fully usable for personal investing:** with priorities 1-4 above → **~2-3 weeks**. Full vision → **~2 months**.

10. PROJECT COMPLETION ESTIMATE

Metric	%
Core intelligence platform (R2 series)	**100%**
Overall feature completion	**70%**
Overall production readiness	**72%**
Estimated project completion	**~74%**

FINAL VERDICT ****GREEN BUILD — GREEN CORE TESTS — STRONG FOUNDATION, CLEAR REMAINING WORK.**** BIST ELITE AI has delivered its defining feature — an Early Opportunity Detection Platform that genuinely composes 11 production engines with deterministic Turkish explanations and zero duplicated scoring logic. The architecture is clean, the reuse culture is real, and the infrastructure story (Docker, systemd, CI/CD, security) is unusually thorough for a project of this size. However: the project ships 5 failing tests, has 5 route collisions, 3 stub/orphan controllers, a missing TradingView provider, an in-memory-only self-learning cycle, no enforced coverage, and ****no live deployment****. The version-control risk is severe (bulk uncommitted). ****Recommendation:**** Treat this as a ****hardening + go-live phase****, not a feature phase. With ~2-3 weeks of focused hardening it becomes genuinely usable for personal investing; the full ML/data-pipeline vision requires ~2 months more.